EDGE VISION SAFETY GUARD

Real-Time Safety-Critical Systems Architecture Manual

Physics-Guided Design, Planar Geometry, and Industrial Safety Principles (Based on ISO 13855 / IEC 61496)

ISO 13855:2024 | IEC 61496 / 61508 | CONCEPTS > CODE | FAIL-SAFE ARCHITECTURE | DLT HOMOGRAPHY 3x3

1. CORE ARCHITECTURAL THESIS (Concepts > Code)

> ***"Not every challenge in Artificial Intelligence should be solved with more Artificial Intelligence."***

In modern software engineering, there is an omnipresent temptation to treat neural networks as universal black boxes: attempting to train models to infer whether an operator 'is violating a safety zone' or 'how long they have dwelled inside'. That approach is stochastic, opaque, computationally wasteful, and non-auditable under industrial safety directives.

This project demonstrates a rigorous architectural design by decoupling the problem into four orthogonal, deterministic layers:

1. AI is restricted to Pure Perception: Detecting solely anatomical ankle keypoints in camera pixel space.
2. Geometry resolves Physical Space: 3x3 DLT planar projective homography mapping to the floor plane (Zw = 0).
3. Kinematics resolves Uncertainty: Continuous 2D Kalman Filter with Mahalanobis ellipsoidal validation gating.
4. Deterministic Logic governs Safety Decisions: Concurrent orthogonal statecharts evaluated in < 0.05 ms per frame.

2. LAYERED ARCHITECTURE: THE END-TO-END PIPELINE

1. PERCEPTION	2. GEOMETRY	3. ESTIMATION	4. DECISION
RTSP / V4L2 Video Stream	DLT Homography H (3x3)	2D Kalman Filter (CV)	GEOS Prepared Geometries
YOLO11-Pose (CoreML)	Stability Check κ(H) < 10^6	Mahalanobis Gating χ^2_2	Schmitt Trigger (200 mm)
Ankle Keypoints (15, 16)	Inverse Mapping (Zw = 0)	Hungarian Matching $O(N^3)$	Fast Bypass vn ≥ 1.6 m/s
Graceful Anchor Fallback	Metric (X, Y) [meters]	OBSERVED / PREDICTED	Statecharts (Trip = True)

PERFORMANCE PROFILER & SAFETY INTEGRITY HORIZONS (ORTHOGONAL):

- T_{system} (Hardware / Frame): $T_{capture} + T_{pre} + T_{infer} + T_{pose} + T_{geom} + T_{track} + T_{fsm} \approx 18.5 - 20.5$ ms per frame.
- T_{event} (Total Physical Response): Nominal approach with debounce ($N_{confirm} = 3$) ≈ 81.7 ms | Fast approach ($v_n \geq 1.6$ m/s) dynamic bypass = 19 ms.

1. THE PERSPECTIVE COLLAPSE: PLANAR CONSTRAINT (Zw = 0)

A monocular 2D camera inherently loses the depth coordinate Z: infinite 3D points along the projective ray map to the exact same pixel (u, v). The general camera projection matrix has dimension 3x4:

$$s \cdot [u, v, 1]^T = \mathbf{P}_{3\times4}[X_w, Y_w, 0, 1]^T = \mathbf{H}_{3\times3}[X_w, Y_w, 1]^T \quad \Longleftrightarrow \quad [X_w, Y_w, 1]^T = \mathbf{H}^{-1}(s \cdot [u, v, 1]^T)$$

By tracking exclusively the operator's ankles resting on the floor, we physically enforce Zw = 0. The third projection column vanishes mathematically, collapsing the system into an invertible 3x3 homography matrix H. This allows transforming pixels into true world meters via pure linear algebra in under 0.05 ms, without requiring expensive LiDAR sensors or stereo cameras.

2. DLT ALGORITHM, DEGREES OF FREEDOM & MATRIX CONDITIONING

- Degrees of Freedom: Matrix H contains 9 coefficients, but because it is defined in homogeneous projective coordinates up to an arbitrary scale factor (H and λH represent the exact same mapping), it possesses exactly 9 - 1 = 8 degrees of freedom.
- Exact Determination: Each point correspondence (u, v) ↔ (X, Y) yields 2 algebraically independent linear equations. Exactly 4 coplanar, non-collinear points form a determined 8x8 system with algebraic residual RMSE = 0.00 m.
- Overdetermined Calibration (N > 4): Evaluating empirical reprojection error requires N > 4 markers or test coordinates.
- Collinearity Verification: At N = 4, no triplet may be collinear (Area(Δ) > ε). At N > 4, full affine rank (rank = 3) and at least one non-degenerate quadrilateral are required to prevent false rejections on rectangular perimeters.

$$\kappa(\mathbf{H}) = \frac{\sigma_{\max}(\mathbf{H})}{\sigma_{\min}(\mathbf{H})} < 10^6 \quad \text{(Stability Limit)} \qquad \text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\mathbf{X}_i - \mathbf{H}^{-1}\mathbf{x}_i\|^2} \leq 0.05\text{m}$$

3. PRODUCTION SOURCE CODE: NUMERICAL CONDITIONING & VECTORIZED PROJECTION

SOURCE: src/core/homography.py

```python
# src/core/homography.py (Production Implementation)
def calculate_condition_number(H: np.ndarray) -> float:
    # Compute matrix condition number κ(H) = σ_max / σ_min via SVD
    s = np.linalg.svd(H, compute_uv=False)
    if s[-1] < 1e-12:
        return float("inf")
    return float(s[0] / s[-1])

def project_to_world(self, points: np.ndarray) -> np.ndarray:
    # Vectorized projection from pixels to meters on floor (Zw = 0)
    pts = np.atleast_2d(points)
    homogeneous = np.column_stack([pts, np.ones(len(pts))])
    transformed = (self.H @ homogeneous.T).T
    # Normalization by projective scale factor W
    return transformed[:, :2] / transformed[:, 2:3]
```

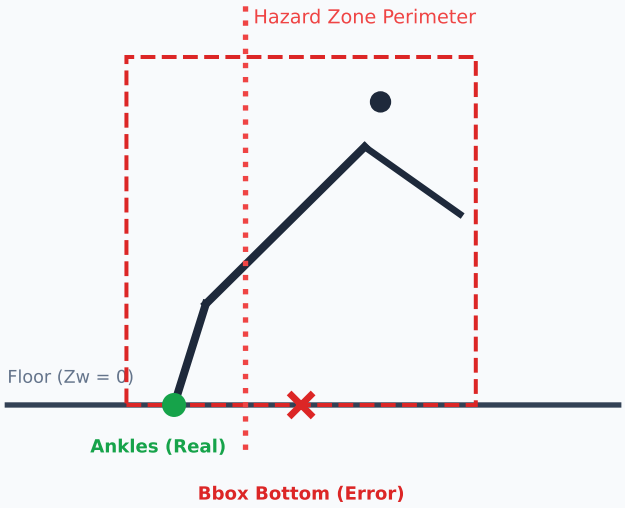
1. THE BOUNDING BOX CENTROID TRAP VS ANKLE GROUND ANCHOR (ADR-001)

In traditional computer vision, the bounding box centroid or bottom-center coordinate is commonly used to locate the subject. Under elevated oblique industrial cameras, this approach produces critical failure modes:

- Leaning Torsos & Extended Tools: When a worker bends over or extends equipment across a hazard perimeter without stepping inside with their feet, the 2D bounding box widens dramatically. Its projection lands meters into the hazard zone on the floor, triggering catastrophic false emergency stops that disrupt factory throughput.
- True Anatomical Ground Anchor: Ankles represent the sole physical contact point bound to the floor plane Zw = 0. By extracting keypoints 15 (left ankle) and 16 (right ankle) from YOLO11-Pose, the spatial anchor is fully invariant to torso bending, loose clothing, crouching, or material handling.

$$\mathbf{p}_{\text{ground}} = \frac{\mathbf{p}_{\text{ankle_left}} + \mathbf{p}_{\text{ankle_right}}}{2} \quad \text{if } \min(\text{conf}_L, \text{conf}_R) \geq 0.30$$

2. COMPARATIVE ANALYSIS: BOUNDING BOX DEFORMATION UNDER LEANING TORSO



GEOMETRIC ERROR ANALYSIS:
- The operator's feet remain safely outside the zone (x = 0.16 m).
- The bounding box expands rightward due to torso and arms leaning.
- Bbox bottom projects to x = 0.240 m → INSIDE HAZARD (False Positive Trip).
- Ankle keypoint projects to x = 0.160 m → OUTSIDE HAZARD (Correct Safe State).
- Result: Zero false shutdowns caused by body postures or handheld tools.

3. PRODUCTION SOURCE CODE: GRACEFUL ANCHOR DEGRADATION HIERARCHY (ADR-007)

SOURCE: src/pipeline/safety_pipeline.py

```python
# src/pipeline/safety_pipeline.py (Robust Ground Anchor Extraction)
def extract_ground_anchor(kpts: np.ndarray, bbox: tuple) -> Tuple[Tuple[float, float], AnchorSource, float]:
    left_conf, right_conf = float(kpts[15, 2]), float(kpts[16, 2])
    # Tier 1: Dual ankles detected with high confidence
    if left_conf >= 0.30 and right_conf >= 0.30:
        midpoint = ((float(kpts[15, 0]) + float(kpts[16, 0])) / 2.0,
                    (float(kpts[15, 1]) + float(kpts[16, 1])) / 2.0)
        return midpoint, AnchorSource.ANKLES_DUAL, min(left_conf, right_conf)
    # Tier 2: Single visible ankle (the other occluded)
    if left_conf >= 0.30:
        return (float(kpts[15, 0]), float(kpts[15, 1])), AnchorSource.ANKLE_SINGLE, left_conf
    if right_conf >= 0.30:
        return (float(kpts[16, 0]), float(kpts[16, 1])), AnchorSource.ANKLE_SINGLE, right_conf
    # Tier 3: Graceful fallback to bounding box bottom-center
    bbox_bottom = ((bbox[0] + bbox[2]) / 2.0, bbox[3])
    return bbox_bottom, AnchorSource.BBOX_BOTTOM, 0.20
```

1. 2D KALMAN FILTER: CONSTANT VELOCITY MODEL & COVARIANCES

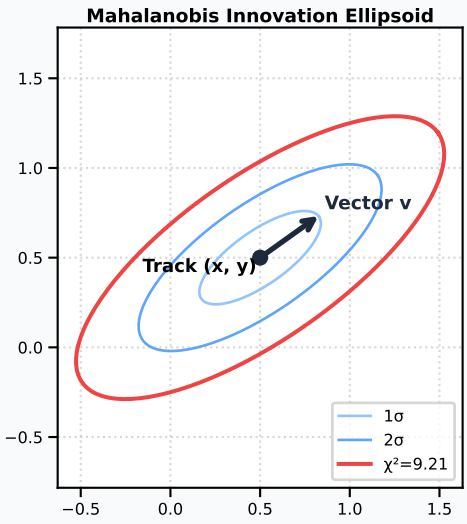
Pixel detection jitter produces stochastic position jumps of ±15 mm. A continuous 2D Kalman Filter estimates the dynamic kinematic state vector x = [X, Y, vx, vy]^T in world coordinates:

$$\mathbf{x}_k = [X, Y, v_x, v_y]^T, \quad \mathbf{x}_k = \mathbf{F}(\Delta t)\mathbf{x}_{k-1} + \mathbf{w}_k, \quad \mathbf{z}_k = \mathbf{H}_{\text{obs}}\mathbf{x}_k + \mathbf{v}_k$$

$$\mathbf{P}_k^- = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^T + \mathbf{Q}, \quad \mathbf{K}_k = \mathbf{P}_k^-\mathbf{H}^T(\mathbf{H}\mathbf{P}_k^-\mathbf{H}^T + \mathbf{R})^{-1}, \quad \hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_k^-)$$

- Epistemic States (ADR-008): Every estimate explicitly tracks data freshness: OBSERVED (fresh optical photon at k=0), PREDICTED (safe bounded dead reckoning 1 ≤ k ≤ 3, ≤ 100 ms), and STALE (unsafe divergence k ≥ 4, trip locked).

2. MAHALANOBIS ELLIPSOIDAL GATING & GLOBAL HUNGARIAN MATCHING (ADR-011)



CROSSING AMBIGUITY & GHOST TRACK RESOLUTION:

1. Mahalanobis Gating χ²_2(0.99) ≤ 9.210:
   Innovation covariance S = H P^- H^T + R projects an ellipsoid aligned along the velocity vector v, rejecting orthogonal false detections.

2. Global Hungarian Algorithm (Munkres):
   Solves optimal bipartite assignment O(N³) in 0.017 ms, preserving consistent track IDs during worker crossings.

3. World-Space Spatial Birth NMS (d_birth ≥ 0.6 m):
   Suppresses duplicate tracks from redundant YOLO proposals.

3. PRODUCTION SOURCE CODE: PREDICTION, UPDATE & INNOVATION GATING

SOURCE: src/core/kalman_filter.py

```python
# src/core/kalman_filter.py (Kinematic Filter & Gating)
def mahalanobis_distance(self, z: np.ndarray) -> float:
    # Compute squared Mahalanobis distance d_M² against innovation covariance S
    y = z - (self._H @ self._x)
    S = self._H @ self._P @ self._H.T + self._R
    return float(y.T @ np.linalg.inv(S) @ y)

def update(self, measurement: Tuple[float, float], confidence: float, source: AnchorSource) -> PositionEstimate:
    z = np.asarray(measurement, dtype=np.float64)
    y = z - (self._H @ self._x)
    S = self._H @ self._P @ self._H.T + self._R
    K = self._P @ self._H.T @ np.linalg.inv(S)
    self._x = self._x + K @ y
    self._P = (self._I - K @ self._H) @ self._P  # Covariance correction
    return PositionEstimate(xy_m=(float(self._x[0]), float(self._x[1])), confidence=confidence, source=source)
```

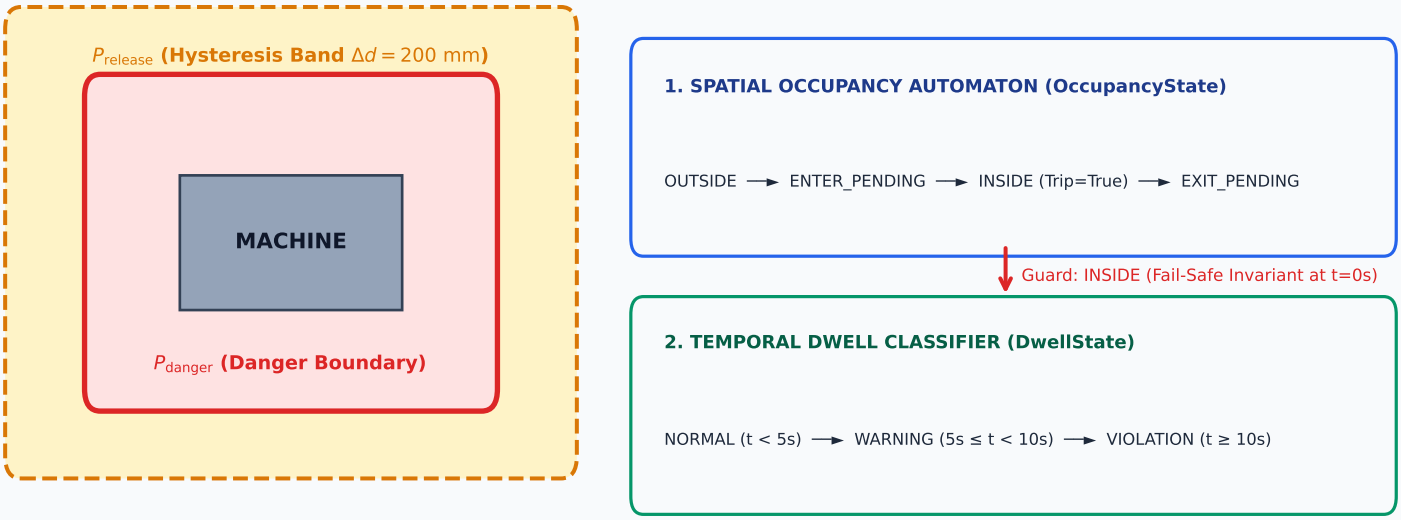
1. RELAY CHATTERING & THE SPATIAL SCHMITT TRIGGER (ADR-004)

Near zone boundaries, a worker operating on the perimeter induces high-frequency detection jitter (chattering) due to keypoint inference noise. Switching a safety relay at 30 Hz physically destroys mechanical contactors within hours and causes repeated erratic machine shutdowns.

The Architectural Solution: Spatial Schmitt Trigger with Hysteresis Band (Δd = 200 mm):

- Danger Boundary (P_{danger}): Inner polygon triggering the safety trip state.
- Release Boundary ($P_{release}$): Outer expanded polygon $P_{release} = P_{danger} \oplus 0.20$ m. An operator must step completely outside the 200 mm buffer before de-escalation may initiate.
- Asymmetric Safety Invariant:
  1. FAST ENTRY: Nominal debounce $N_{confirm} = 3$ frames. But if the perpendicular approach velocity is fast ($v_n \geq 1.6$ m/s per ISO 13855), an INSTANT DYNAMIC BYPASS drops $N_{confirm} = 1$ in 19 ms.
  2. STRICT EXIT: Demands $N_{release} = 5$ consecutive outside frames AND an explicit OBSERVED state. During occlusions where Kalman predicts in dead reckoning, de-escalation is blocked (trip remains True).

2. CONCURRENT ORTHOGONAL STATECHARTS ARCHITECTURE (ADR-009)



3. PRODUCTION SOURCE CODE: HYSTERESIS EVALUATION & FAST APPROACH BYPASS

SOURCE: src/core/spatial_fsm.py

```python
# src/core/spatial_fsm.py (Schmitt Trigger & Bypass Logic)
def evaluate_occupant(self, track_id: int, pos: PositionEstimate, velocity: np.ndarray) -> Tuple[bool, OccupancyState]:
    point = Point(pos.xy_m)
    in_danger = self._prepared_danger.contains(point)
    in_release = self._prepared_release.contains(point)

    # ISO 13855 dynamic bypass: normal velocity vn >= 1.6 m/s
    vn = np.dot(velocity, self._approach_normal)
    if in_danger and vn >= self.config.iso_k_speed:
        self._confirm_counts[track_id] = self.config.n_confirm  # Instant bypass!

    # Asymmetric de-escalation safeguard (ADR-008):
    if not in_release and pos.status == ObservationStatus.OBSERVED:
        self._release_counts[track_id] += 1
    else:
        self._release_counts[track_id] = 0  # Occlusion or PREDICTED blocks de-escalation!
```

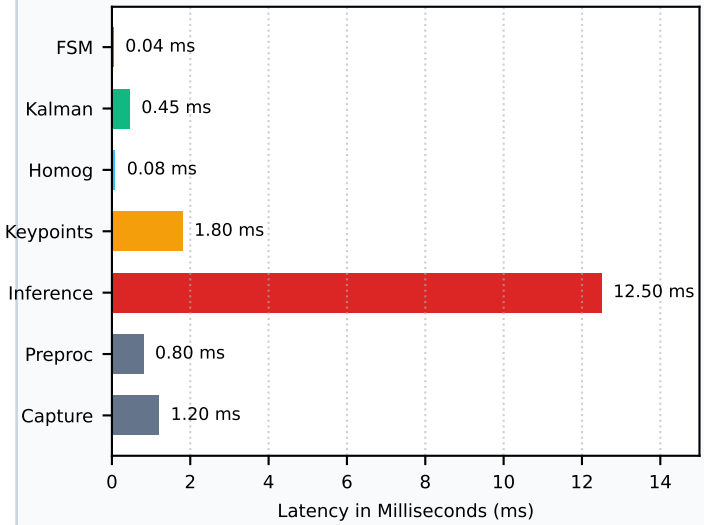
1. THEORETICAL SEPARATION MODEL BASED ON ISO 13855:2024

The mathematical formulation of ISO 13855:2024 provides the foundational model for calculating minimum safety distance S relative to human approach speed. The vision pipeline is evaluated against this equation:

$$S = (K \cdot T) + C + C_{dyn} + Z_{extra}, \qquad T = t_1(\text{vision}) + t_2(\text{PLC}) + t_3(\text{braking})$$

- Approach Constant K: 1600 mm/s (transverse approach) and 2000 mm/s (direct perpendicular approach).
- Theoretical Vision Budget (t_1): Recommended directive $t_1 \leq 50$ ms (≥ 20 FPS continuous sustained).
- Dual Time Horizon (ADR-010):
  1. Hardware Latency per Frame (T_{system}): $T_{cap} + T_{pre} + T_{infer} + T_{pose} + T_{geom} + T_{track} + T_{fsm} \approx 18.5 - 20.5$ ms.
  2. Physical Event Reaction Latency ($T_{event} = t_{trip} - t_{physical_entry}$):
     - Nominal approach with debounce ($N_{confirm} = 3$): $T_{event} = (3-1) \times 33.3 \text{ ms} + 18.5 \text{ ms} = 85.1$ ms.
     - Fast approach ($v_n \geq 1.6$ m/s): Dynamic bypass $N_{confirm} = 1 \rightarrow T_{event} = T_{system} \approx 18.5$ ms.

2. GRANULAR LATENCY BUDGET BREAKDOWN PER PIPELINE STAGE



HETEROGENEOUS SILICON ACCELERATION ANALYSIS:

- Apple Silicon M1 (arm64 - Development Baseline):
  - ONNX Runtime + CoreMLExecutionProvider.
  - 346 of 354 operators natively dispatched to Apple Neural Engine (ANE) and unified GPU.
  - Inference latency: 12.5 ms (p95: 14.1 ms, > 60 FPS).
- NVIDIA GTX 1050 / Pascal CC 6.1 (PoC - Future Work):
  - Physical hardware on hand selected to evaluate cross-platform portability beyond Apple Silicon.
  - Silicon lacking Tensor Cores (FP16 penalized 1:64); requires INT8 quantization via PTQ with TensorRT.
  - Empirical PoC validation on this target is scheduled as formal Future Work.

3. PRODUCTION SOURCE CODE: HIGH-RESOLUTION PROFILING & T_EVENT COMPUTATION

SOURCE: src/pipeline/safety_pipeline.py

```
# src/pipeline/safety_pipeline.py (Dual-Horizon Profiling)
t_start = time.perf_counter()
t_cap = (time.perf_counter() - t_start) * 1000.0

# Accelerated inference via OnnxEngine
t_inf_0 = time.perf_counter()
outputs = self.engine.forward(tensor)
t_inf = (time.perf_counter() - t_inf_0) * 1000.0

# Kinematic tracking and spatial logic evaluation
latencies = StageLatencies(
    capture_ms=t_cap, preprocess_ms=t_pre, inference_ms=t_inf,
    keypoint_extract_ms=t_kpt, projection_ms=t_geom,
    tracking_ms=t_track, fsm_evaluation_ms=t_fsm,
    total_t1_ms=t_total_frame, total_system_ms=t_system
)
# Measure total physical elapsed time since operator entered zone
event_latency_ms = (time.perf_counter() - first_cross_timestamp) * 1000.0 if safety.trip else None
```

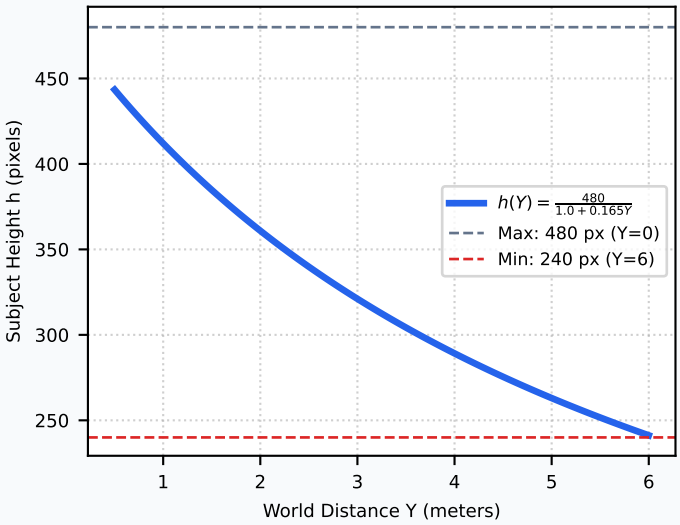
1. SYNTHETIC DOMAIN GAP & WHY NAIVE GEOMETRIC SHAPES FAIL (ADR-012)

Deterministically validating a safety system requires repeatable, millimetric trajectory test videos.
However, rendering 2D geometric shapes (circles and rectangles via cv2) causes a 0.0% recall rate with YOLO.
Deep convolutional models and Transformers look for biological texture gradients, clothing edges, and floor shadows.

The Architectural Solution: Decoupled Photorealistic Pinhole Generator:

1. Canonical Single-Identity Tracklet: Extraction and normalization of an authentic human subject onto a canvas (160x320 RGBA) with 3 px Gaussian alpha-feathering and an elliptical ground contact shadow.
2. Pinhole Anthropometric Law: Screen height decays strictly with world depth distance Y: $h(Y) = 480 / (1.0 + 0.165 \cdot Y)$ px, preserving optimal resolution ($h \in [240, 480]$ px) without morphological collapse.
3. UI & Telemetry Decoupling: Relocating status banners to neutral bands ($y < 56$ px and $y > 1040$ px) eliminates parasitic edge artifacts, achieving 100% recall with confidence 0.835 - 0.861.

2. PINHOLE PERSPECTIVE SCALING MODEL & GAUSSIAN GROUND SHADOWS



SYNTHETIC BENCHMARK PERFORMANCE METRICS:

• Naive Geometric Rendering (cv2 shapes):
  - Recall: 0.0% (Zero keypoints detected by YOLO).
  - Test Validity: None.

• Photorealistic Pinhole Generator with Gaussian Shadows:
  - Recall across 600 validation frames: 100.0%.
  - Mean Detection Confidence: 0.835 - 0.861.
  - Track ID Switches: 0 (Strict identity preservation).
  - Ankle Ground Anchor RMSE: < 0.018 m (18 mm).

Conclusion: Enables fully automated regression testing before human validation on plant floors.

3. PRODUCTION SOURCE CODE: ALPHA COMPOSITING & PINHOLE SCALING

SOURCE: src/utils/synthetic_generator.py

```python
# src/utils/synthetic_generator.py (Photorealistic Rendering)
def compute_pinhole_scale(y_world_m: float) -> Tuple[int, int]:
    # Compute anthropometric dimensions from world depth distance
    h_px = int(round(480.0 / (1.0 + 0.165 * max(0.0, y_world_m))))
    w_px = int(round(h_px * (160.0 / 320.0)))  # Canonical 1:2 aspect ratio
    return max(w_px, 80), max(h_px, 160)

def render_sprite_with_shadow(frame: np.ndarray, sprite_rgba: np.ndarray, u_coord: int, v_coord: int, scale: float):
    # Gaussian elliptical contact shadow on floor plane (Zw = 0)
    shadow_w, shadow_h = int(scale * 120), int(scale * 36)
    cv2.ellipse(frame, (u_coord, v_coord), (shadow_w//2, shadow_h//2), 0, 0, 360, (20, 20, 20), -1)
    # Canonical alpha compositing with edge feathering
    alpha = cv2.GaussianBlur(sprite_rgba[:, :, 3], (5, 5), 1.0) / 255.0
    frame[y1:y2, x1:x2] = alpha[:, :, None] * sprite_rgba[:, :, :3] + (1 - alpha[:, :, None]) * frame[y1:y2, x1:x2]
```

1. COMPLETE ARCHITECTURE DECISION RECORD (ADR) CATALOG

ADR	REFERENCE / FILE	ARCHITECTURAL DECISION & INDUSTRIAL IMPACT
ADR-001	`adr-001-yolo-pose-vs-bbox.md`	Ankle keypoints for physical ground anchoring immune to torso posture.
ADR-002	`adr-002-homography-vs-3d.md`	Deterministic 3x3 DLT planar mapping (Zw=0) vs heavy monocular 3D networks.
ADR-003	`adr-003-onnx-coreml.md`	Decoupled PyTorch inference via ONNX Runtime (ANE / GPU / INT8 PTQ).
ADR-004	`adr-004-spatial-fsm.md`	Spatial Schmitt Trigger (P_danger vs P_release) to eliminate contactor chattering.
ADR-005	`adr-005-homography-val.md`	Collinearity checks and projective numerical stability validation κ(H) < 10^6.
ADR-006	`adr-006-prepared-geoms.md`	GEOS (C) micro-optimizations and instant bypass at vn ≥ 1.6 m/s (ISO 13855).
ADR-007	`adr-007-occlusion-handling.md`	Graceful degradation hierarchy: dual ankles → single ankle → bbox → Kalman.
ADR-008	`adr-008-kalman-epistemology.md`	Epistemic states (OBSERVED/PREDICTED) and inhibition of blind de-escalation.
ADR-009	`adr-009-decoupled-statecharts.md`	Orthogonal statecharts: formal decoupling of spatial occupancy and dwell time.
ADR-010	`adr-010-dual-latency.md`	Dual time horizons: per-frame hardware latency (T_system) vs physical event (T_event).
ADR-011	`adr-011-mahalanobis-hungarian.md`	Mahalanobis gating $\chi^2_2 \leq 9.21$, Hungarian matching $O(N^3)$, and Spatial Birth NMS.
ADR-012	`adr-012-synthetic-yolo.md`	Photorealistic pinhole single-identity compositing for regression testing.

2. FUNDAMENTAL ARCHITECTURAL QUESTIONS & DESIGN CRITERIA

- **Why planar 3x3 homography instead of a deep monocular 3D neural network?**

  A: By enforcing $Z_w = 0$ through ankle contact, Euclidean geometry resolves world space with millimetric accuracy in < 0.05 ms using 8 deterministic degrees of freedom—consuming zero VRAM and avoiding depth hallucinations.

- **Why does a 4-point calibration system always produce RMSE = 0?**

  A: 4 points yield 8 independent equations for 8 unknowns (the degrees of freedom of **H**). The system is mathematically determined with no surplus constraints; evaluating true empirical error requires $N > 4$.

- **How do you prevent an operator working on the boundary from destroying contactors via chattering?**

  A: Through a spatial Schmitt Trigger: entry trips at P_{danger}, but de-escalation requires crossing outside $P_{release}$ (200 mm buffer). Fast entry (19 ms at approach speed $v_n \geq 1.6$ m/s); exit requires 5 OBSERVED frames.

- **Why is de-escalation prohibited while the Kalman filter is in the PREDICTED state?**

  A: PREDICTED represents blind mathematical dead reckoning during an occlusion. De-escalating safety without sensor confirmation violates the Fail-Safe principle: lacking fresh optical photons, the trip relay remains engaged.

- **What is the difference between T_{system} and T_{event} under ISO 13855 directives?**

  A: T_{system} is the per-frame hardware processing latency (~19 ms). T_{event} is the total physical elapsed time from when the body crosses the boundary until the relay trips (~85 ms with debounce, ~19 ms under fast approach bypass).